Task 3: Evidence Handling

1. Introduction

Digital evidence plays a critical role in modern investigations involving cybercrime, fraud, and data breaches. Unlike physical evidence, digital evidence is highly volatile and can be easily altered, deleted, or corrupted if not handled properly. Therefore, forensic investigators must follow strict protocols to ensure that evidence remains authentic, intact, and legally admissible.

Evidence handling involves a series of controlled procedures including identification, acquisition, preservation, transportation, and storage. This report explores these concepts in detail and demonstrates a practical implementation using cryptographic hashing.

2. Objectives

The primary objectives of this task are:

- To understand the **principles of digital evidence handling**
- To learn **forensic acquisition techniques**
- To ensure **data integrity through hashing algorithms**
- To understand **write protection mechanisms**
- To implement a **practical integrity verification process**
- To maintain proper **documentation and audit trail**

3. Digital Evidence Handling Lifecycle

Digital evidence handling follows a structured lifecycle:

1. Identification
2. Collection
3. Acquisition
4. Preservation
5. Transportation
6. Storage
7. Analysis

Each stage must be carefully executed to avoid contamination.

4. Key Concepts in Evidence Handling

4.1 Write Blockers (Write Protection Mechanism)

A write blocker is a critical forensic tool used to prevent any modification to the original storage media.

Technical Role:

- Blocks write commands at hardware/software level
- Allows read-only access to evidence

Types:

- Hardware Write Blockers (USB/SATA devices)
- Software Write Blockers

Importance:

- Prevents OS from altering metadata (timestamps, logs)
- Ensures forensic soundness

4.2 Forensic Imaging

Forensic imaging is the process of creating an exact bit-by-bit copy of a storage device.

Types of Images:

- Physical Image (entire disk)
- Logical Image (specific files/folders)

Tools Used:

- FTK Imager

Why Important:

- Analysis is performed on the copy, not original
- Preserves original evidence

4.3 Hashing (Cryptographic Integrity Verification)

Hashing is used to verify that evidence has not been altered.

Common Algorithms:

- MD5 (fast but less secure)

- SHA-1 (deprecated in some cases)
- SHA-256 (recommended standard)

Hash is like a **digital fingerprint**

Working Principle:

- Input → Hash Function → Fixed-length output

If even 1 bit changes → completely different hash

4.4 Evidence Storage and Security

Proper storage is essential to maintain long-term integrity.

Best Practices:

- Use anti-static and tamper-proof bags
- Store in access-controlled environment
- Maintain access logs
- Use encryption if required

5. Evidence Handling Workflow

Step 1: Identify evidence source (file/device)
Step 2: Secure the evidence (avoid direct access)
Step 3: Apply write blocker (if device-based evidence)
Step 4: Create forensic image (optional but recommended)
Step 5: Generate initial hash (SHA-256)
Step 6: Perform analysis on copied data
Step 7: Generate final hash
Step 8: Compare hashes for integrity verification
Step 9: Store evidence securely with proper logs

6. Practical Implementation

Case Title:

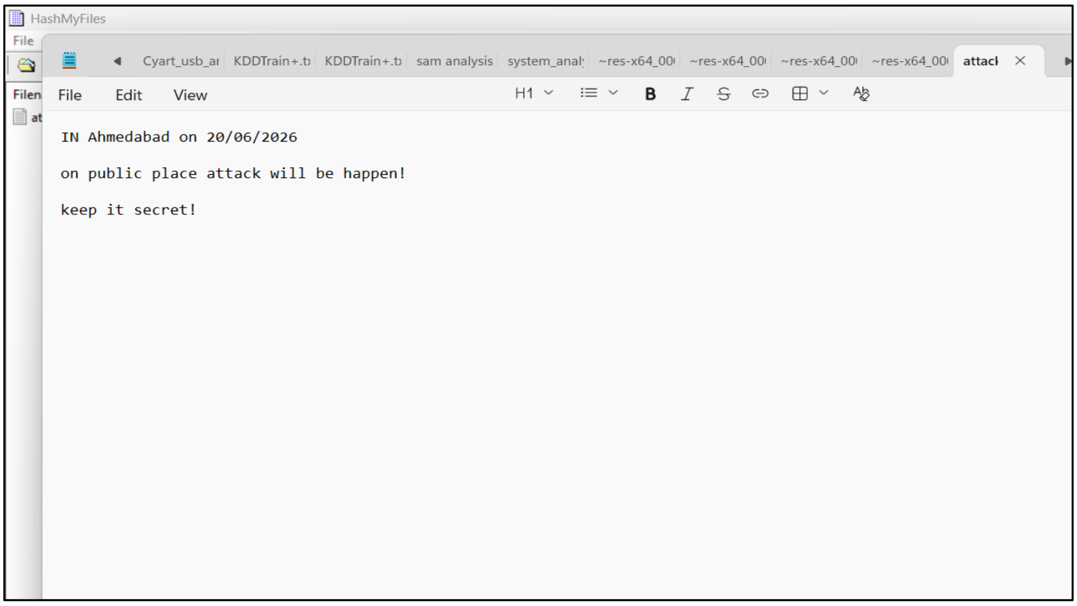
Verification of Evidence Integrity Using Hashing

Case Description:

A digital file is treated as forensic evidence. The objective is to verify that the file remains unchanged during the investigation process using SHA-256 hashing.

Step 1: Evidence Identification

- Selected file: Sample document (PDF/DOCX)
- Treated as digital evidence



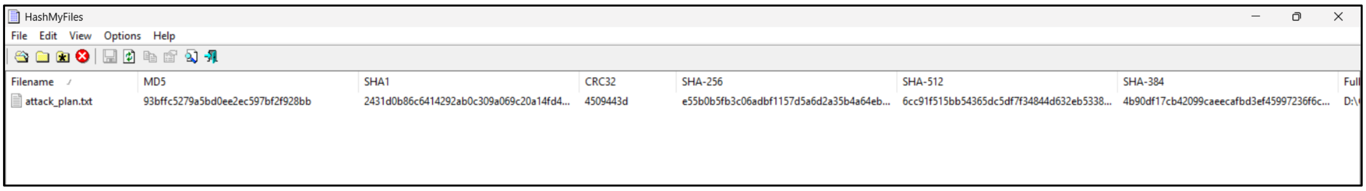
Find a txt file from the attacker computer

Step 2: Initial Hash Generation

- Tool used: HashMyFiles

Generated SHA-256 Hash:

e55b0b5fb3c06adbf1157d5a6d2a35b4a64eb79a272a093756292775eaf382a7



Step 3: Controlled Analysis

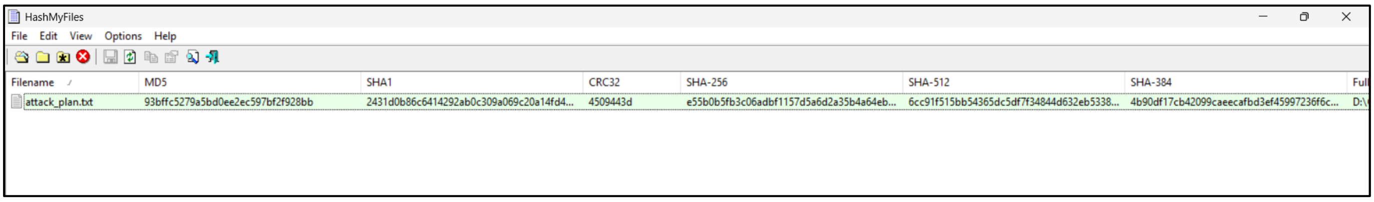
- File opened in read-only mode
- Content reviewed
- No modification performed

Step 4: Final Hash Generation

- Hash generated again using same tool

Generated SHA-256 Hash:

e55b0b5fb3c06adbf1157d5a6d2a35b4a64eb79a272a093756292775eaf382a7



Step 5: Hash Comparison

- Initial Hash = Final Hash

Result: Integrity maintained
If mismatch → Evidence compromised

7. Results and Observations

- Hash values before and after analysis were identical
- No evidence of tampering detected
- Confirms reliability of forensic handling

8. Tools Used

- HashMyFiles – Hash generation

9. Challenges in Evidence Handling

- Risk of accidental data modification
- Lack of proper forensic tools
- Improper documentation
- Human error during handling

10. Best Practices

- Always work on a forensic copy, not original
- Use write blockers for physical devices
- Generate hash before and after analysis
- Maintain proper chain of custody
- Store evidence securely

11. Importance in Legal Context

Proper evidence handling ensures:

- **Admissibility in court**
- Protection against defense claims
- Credibility of investigation
- Compliance with forensic standards

12. Conclusion

Evidence handling is a foundational aspect of digital forensics. Proper techniques such as write blocking, forensic imaging, and hashing ensure that digital evidence remains intact and trustworthy. This task demonstrated the importance of maintaining integrity through practical hashing, reinforcing the need for strict adherence to forensic procedures.